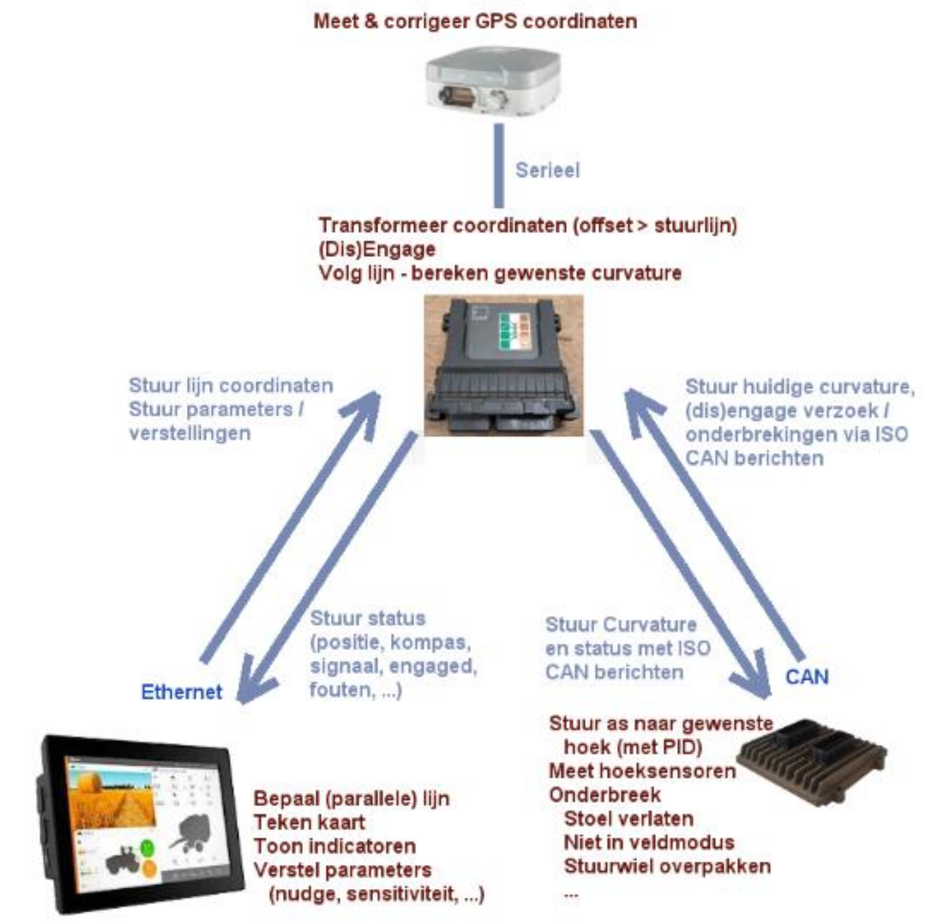
1. AutoSteer system overview

GPH currently has the following system (optionally) installed. The “Autonomous Driving Computer” would be fifth component, which basically takes over the Line Management partially from the Display.



- **TOPCON AGS-2 Receiver:**
  - Measures GPS & corrects with PPP (Sattelite) to ~4..10 cm precision)
  - Coordinate = the receiver itself, no offset applied yet
- **AgJunction ECU-S1:**
  - Reads GPS coordinates from Receiver via RS232 NMEA message
  - Transforms GPS coordinates to machine pivot point & header, applying tilt-correction from an IMU sensor as well
  - Receives line coordinates from the Display to follow (via Ethernet)
  - Receives (dis)Engage command & steeraxle sensor feedback from IQAN (via CAN)
  - Instructs the IQAN how to put the wheels; i.e. the steering command
- **OXBO Cab Display:**
  - Does the Line Management (create AB line & its parallels, import line from external source, ...)
  - Draws a map & keeps track of where the machine has been
  - Shows some alerts & indicators (signal ok, steering interrupted because XYZ)
  - Can tweak a few (steering aggression) parameters on the AgJunction
- **OXBO MC42-Propel:**
  - Actuates the steering valve (set to X degrees according to desired curvature from SteerController)
  - Delivers feedback (actual angle) to SteerController
  - Sends (dis)Engage commands to SteerController

2. Message Overview & Sequence

BUS	SendRate	PGN		PRIO	DLC	Category	Message Name	Abbreviation	Sender	Receiver
Main Bus	100 ms or on state change	FFBE	65470	3	8	Obstacle Detection	Obstacle Detection State and Zones	ODSZ	Obstacle Detector	IQAN MC4x Propel
Main Bus		FFBF	65471	4	8	Obstacle Detection	reserved - not used (yet)			
?	500 ms or on state change	FFC0	65472	4	8	Weed_Trash Detection	Weed & Trash Detection State and Result	WTDSR	Trash Detector	Display
Main Bus		FFC1	65473			Weed_Trash Detection	reserved - not used (yet)			
Displ. 2	100 ms	FEF3	65267	6	8	GPS	Vehicle Position 1 (J1939 standard msg)	VP1	Display	AutoDrive
Displ. 2	200 ms	FEE8	65256	6	8	GPS	Vehicle Direction and Speed	VDS	Display	AutoDrive
Displ. 2	200 ms	FFCA	65482	6	8	Autonomous Driving	DirectSteer Status	DSSTAT	Display	AutoDrive
Displ. 2	5000 ms or onChange	FFCB	65483	6	8	Autonomous Driving	Field Anchor Point	DSAP	Display	AutoDrive
Displ. 2	1000 ms or onChange	FFCC	65484	6	8	Autonomous Driving	AutoDrive Job	ADJOB	AutoDrive	Display
Displ. 2	streaming *	FFCD	65485	6	8	Autonomous Driving	AutoDrive WayPoint info	ADWPI	AutoDrive	Display
Displ. 2	1000 ms	FFD0	65488	6	5	Tenderness	Crop/Soil Analysis Value (Tenderness)	CSAVAL	Tenderness Unit	Display
Displ. 2	3000 ms or onChange	FFD1	65489	6		Tenderness	Crop/Soil Analysis Sensor Unit Status	CSASUS	Tenderness Unit	Display
Displ. 2	3000 ms or onChange	FFD2	65490	6		Tenderness	Crop/Soil Analysis Sensor Unit Control	CSAUCT	Display	Tenderness Unit
Displ. 2		FFDA	65498	6	8	PeaSense	PeaSense Function Settings	PSFS	Display	PeaSense
		FFDB	65499			PeaSense	reserved - not used (yet)			
Displ. 2	When needed	FFDC	65500	6	8	PeaSense	Automatic Control Adjustment Request	ACAR	PeaSense	Display
Displ. 2	3000 ms or onChange	FFDD	65501	6	3	PeaSense	PeaSense Overall Settings	PSOS	Display	PeaSense

Event	AutoDrive System message		Display message
Machine in field, not started	ADJOB (systemActive false)	→	
Wait for GPS PPP signal...		←	VP1 & VDS (GPS coords / speed / compass) DSSTAT (GPS correction available)
Line info received, activate system	ADJOB (systemActive true)	→	
Compute anchor point		←	DSAP (Anchor point)
Stream (1st batch) waypoints	ADWPI	→	
Command forward	ADJOB (systemActive + RUN)	→	
Report status		←	VP1 & VDS (GPS coords / speed / compass) DSSTAT (Autosteer engaged y/n)
Report status	ADJOB (waypoint index)	→	
Stream (next batch) waypoints	ADWPI	→	

3. GPS coordinates

The display will constantly forward the (transformed) GPS coordinates, using these 2 (SAE standard) messages – also when the AutoDrive system is not active. Note that the GPS coordinate presents the header-center at ground-level, which is not neccesarily the machine center if the head has an offset.

VP1	**Vehicle Position 1**							
(J1939) Message from Display to AutoDrive System, containing Header center/ground GPS position								
	Standard SAE message							
PGN	65501	0xFEF3			**CAN BUS**	bus 2 on Main Display		
source	40				**Priority**	6		
Transmit ra	Every 100 msec				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1		Latitude						
Byte2				Offset = -210 degrees				
Byte3				Scale = 0,0000001 deg/bit				
Byte4								
Byte5		Longitude						
Byte6				Offset = -210 degrees				
Byte7				Scale = 0,0000001 deg/bit				
Byte8								

VDS	**Vehicle Direction and Speed**							
(J1939) Message from Display to AutoDrive System, containing additional (GPS) info								
	Standard SAE message							
PGN	65256	0xFEE8			**CAN BUS**	bus 2 on Main Display		
source	40				**Priority**	6		
Transmit ra	Every 200 msec				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1		Compass Bearing		1/128 deg/bit				
Byte2				no offset				
Byte3		GroundSpeed KPH		1/256 km/h per bit				
Byte4				no offset				
Byte5		Pitch angle (deg)		1/128 deg/bit				
Byte6				offset -200 degrees				
Byte7		Altitude mtr		0.125 m/bit				
Byte8				offset -2500 mtr				

If the signal drops, the Latitude & Longitude will be set the 0xFFFFFFFF (not available). Note that the compass bearing might be wrong or undetermined when when the machine didn't move yet.

4. Job Start

Normally, DirectSteer would start a new job either manually or when entering another field (geofence). Starting a new job would:

- Clear the coverage map (display keeps draws where the machine has been)
- Resets lines
- Compute an anchor point (typically the field center or the current machine point)
    - This anchor will be used to communicate coordinates in a local system, centimeters north / east relative to anchor

Since this trigger may not be available now, the AutoDrive system will trigger instead. To do so,

1. The AutoDrive system first has to await a valid (corrected) GPS signal, which may take a few minutes after the machine (re)starts.
2. Machine must allow AutoDrive (fieldMode & option enabled by operator / machine)
3. The machine must be inside the field
4. Waypoints / line available

DSSTAT	**DirectSteer Status**							
(J1939) Message from Display to AutoDrive System, containing current state								
PGN	65482	0xFFCA			**CAN BUS**	bus 2 on Main Display		
source	40				**Priority**	6		
Transmit ra	Every 200 msec				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1	GPS PPP available		AutoSteer Engaged		Header Down		Current Direction	
Byte2	AutoSteer interrupt/reject reason						AutoDrive allowed	
Byte3		Perpendicular distance to line cm					1 cm per bit	
Byte4							offset -1000 cm	
Byte5		Overlap setting cm					offset -125 cm	
Byte6		Picker Reel width (shaft-to-shaft) cm					1 cm per bit	
Byte7		lhTipDistance cm					no offset	
Byte8					1 cm per bit, no offset			

ADJOB	**AutoDrive Job**							
(J1939) Message AutoDrive System to Display, containing job state								
PGN	65484	0xFFCC			**CAN BUS**	bus 2 on Main Display		
source	29 ?				**Priority**	6		
Transmit ra	Every 1s or when state changed				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1								
Byte2		Error Code (0 = ok)			RunCommand		SystemActive	
Byte3		Current waypoint index (i.e. #1403 out of #5000) - i.e. progress						
Byte4								
Byte5		Line total point count						
Byte6								
Byte7		Job ID (0..65530)						
Byte8								

(left) DirectSteer status (from Display). AutoDrive should check the GPS PPP available and AutoDrive allowed bits.
(right) AutoDrive Job. Turn on the “systemActive” bit once the conditions are met (do NOT turn on RunCOmmand yet). If the AutoDrive system has a job ID, it can be communicated as well with the last 16 bits.

If the SystemActive bits toggles on, the DirectSteer system will start a new job. This includes computing a (new) anchor point. Waypoint coordintes that will be streamed are ought to be relative to be relative to this anchor. This allows a more compact & easier to handle coördinate system.

DSAP	**DirectSteer Anchor Point**							
(J1939) Message from Display to AutoDrive System, containg Job/Field anchor point								
PGN	65483	0xFFCB			**CAN BUS**	bus 2 on Main Display		
source	40				**Priority**	6		
Transmit ra	Every 5s or when changed				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1		Anchor Latitude						
Byte2				Offset = -210 degrees				
Byte3				Scale = 0,0000001 deg/bit				
Byte4								
Byte5		Anchor Longitude						
Byte6				Offset = -210 degrees				
Byte7				Scale = 0,0000001 deg/bit				
Byte8								

Be aware that 2 different machines may have 2 different anchor points!

As long as the system is inactive, anchor longitude and latitude are set to 0xFFFFFFFF

Once the anchor point has been received, the AutoDrive unit can start streaming waypoints. Bare in mind the “RunCommand” from the ADJOB message should remain off, as the display still doesn’t know (all) required waypoints yet.

5. Streaming Waypoints towards the Display

Once the anchor point has been set, the AutoDrive system can translate absolute GPS coordinates (degrees) to local-system coordinates, which are centimeters offset relative to the anchor. For example,

{-51050 , +912 }   means 510,5 meters west from the anchor, 9,12 meters north from the anchor

ADWPI - AutoDrive WayPoint Info								
(J1939) Message AutoDrive System to Display, containing current status								
PGN	65485	0xFFCD			**CAN BUS**	bus 2 on Main Display		
source	29 ?				**Priority**	6		
Transmit rate		See notes			**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1		Point index		0 = first point				
Byte2								
Byte3		Point coordinate East to Anchor				1 cm per bit		
Byte4						Offset -250000 cm (-25 km)		
Byte5								
Byte6		Point coordinate North to Anchor				1 cm per bit		
Byte7						Offset -250000 cm (-25 km)		
Byte8		Reserved			is Reverse point		is Headland point	

A CAN message can only contain 1 point at a time, so an index number is given to tell which waypoint this is. Besides coordinates, it can carry additional information:

- Is Headland point   meaning the header should be lifted, and typically a (sharp) turn is coming so reduce the speed as well).
- Is Reverse point   meaning the machine has to travel in reverse to reach this point)
- Reserved   for future usage (speedlimit? Additional steer instructions?)

Index[0] means the very first waypoint of the entire line. If the line has 6000 points in total (as communicated in the ADJOB message), Index[5999] would present the last point.

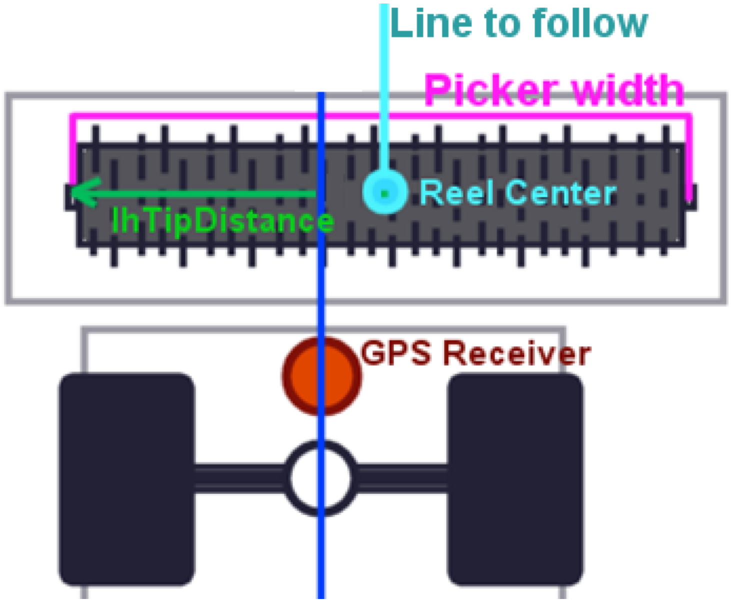
- The system is allowed to stream ALL points, though it's is also allowed to only stream 100 points at a time.
  - So when passing point 75, stream the next 100 (see chapter 5.2)
  - Or just stream the entire line if that is easier (the display will draw the entire line as well then). Be aware this takes some time though!

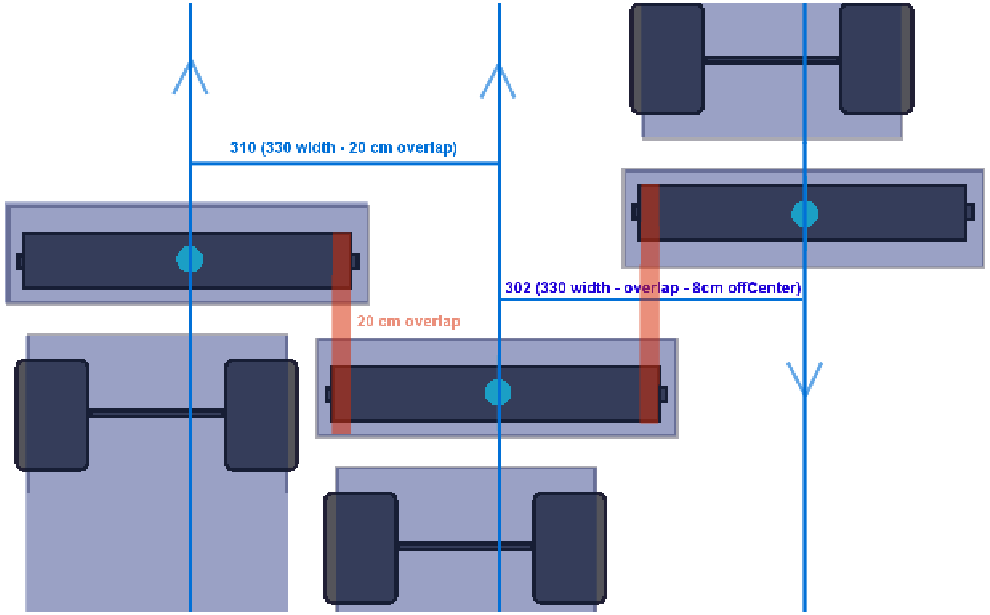
- Streaming pace;
  - When streaming 100 points, PGN 0xFFCD needs to be sent 100 times as well, each time with different point index
  - Since only a relative small amount of CAN messages can be committed simultanously, it is recommended to pause 10 milliseconds after each message
    - So sending 100 points would require 1 second
  - Or if we prefer to send the entire line at once, say 20k points, it would take (20k / 100) = 200 seconds
  - If this is too slow; it is allowed to send the same PGN multiple times simultanously. For example:
    - 0 ms - send point1 & 2 10 ms – send point 3 & 4 20 ms – send point 5 & 6 ....
    - This would reduce the streaming time by 50%
  - The system can engage when at least the first 100 points have been streamed (and it can keep streaming after that if preferred).

- If the line changes on the fly, KEEP the passed points. For example:
  - Machine just passed waypoint #2000
  - The line suddenly changes
  - Point 0..2000 should remain intact, as they were given before. Points 2001+ can be changed
  - It is allowed to increase or reduce the point count'
  - This means passed points don't have to be re-communicated

- AgJunction curve path requirements:
  - AgJunction will (re)smooth the line for its own usage, nonetheless it is recommended to ensure Spacing and Angle limits as described in chapter 5.2

The AutoSteer system will try to put the header-center on top of the steerline. Be aware that for machines with an excentric header position, the distancing between lines varies depending from which direction the machine is coming:





Overlap, reel width (that is the actual picking reel inside (shaft-to-shaft), not the whole header) & lhTipDistance are broadcasted in the DSSTAT message:

DSSTAT	**DirectSteer Status**							
(J1939) Message from Display to AutoDrive System, containing current state								
PGN	65482	0xFFCA			**CAN BUS**	bus 2 on Main Display		
source	40				**Priority**	6		
Transmit ra	Every 200 msec				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1	GPS PPP available		AutoSteer Engaged		Header Down		Current Direction	
Byte2	AutoSteer interrupt/reject reason						AutoDrive allowed	
Byte3		Perpendicular distance to line cm					1 cm per bit	
Byte4							offset -1000 cm	
Byte5		Overlap setting cm					offset -125 cm	
Byte6		Picker Reel width (shaft-to-shaft) cm					1 cm per bit	
Byte7		lhTipDistance cm					no offset	
Byte8					1 cm per bit, no offset			

Overlap should be set BEFORE the job starts (typically a fixed setting at 18..25 cm). It cannot be changed on the fly, as the whole path would require recomputation.

The reel width is a constant value, depending on what type of header is mounted.

The lhTipDistance can vary, if the head was shifted. It is the distance between the left reel shaft and the SteerLine (see picture above). So if the head was shifted to the right, the lhTipDistance becomes smaller than half of the reel width. Shifting does not affect where the lines should be, as the machine will adapt on the fly. Nonetheless the info is shared.

5.2 Streaming Linepoints towards the AgJunction

The AgJunction should(*) be able to follow a curved line, though it has certain requirements:

6.1.2 Curve Path requirements

The Steering system can accept a curved path of up to 100 points.

Point Spacing:
Minimum: 0.3m
Maximum: 4.5m
Maximum delta angle change between 2 segments: 30 degrees

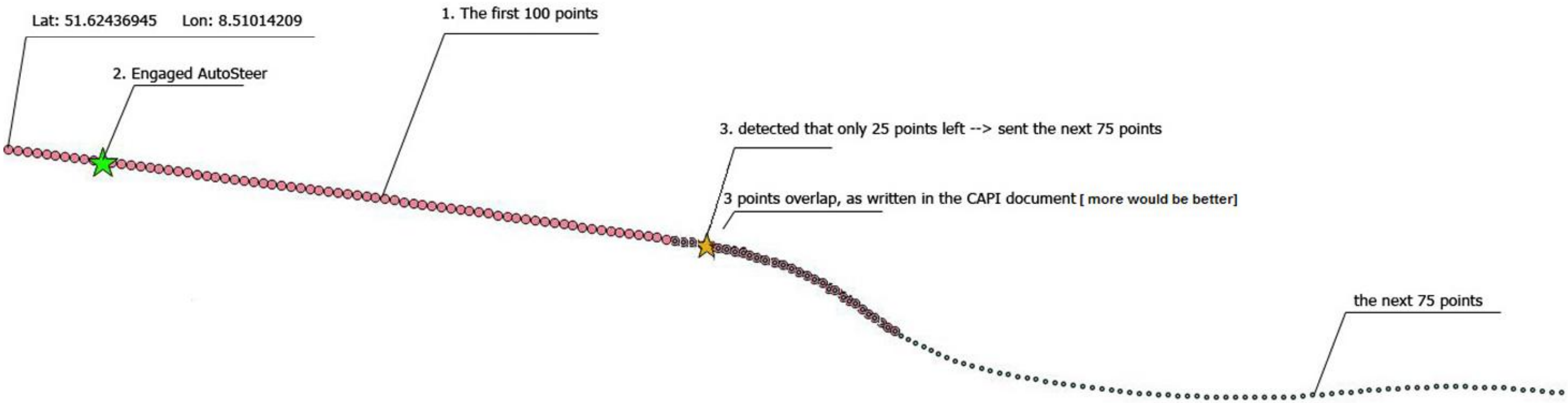
Both the Display and AgJunction itself will smooth / rebuild the line, but it's probably good if the AutoDrive ECU uses the same logic.

* We noticed the machine having difficulties with following a curved line, eventually losing track and quitting autosteer as the AgJunction missed too many waypoints. Possible causes could be:
- Too much "zig-zagging" on a relative short distance (we never used very sharp turns)
- Wheels not steering quick enough (aggressive) enough on the propel (we can tweak that)
- Streaming logic not properly implemented on the Display (see below)

As curves are rarely used in normal use cases, we never put much effort in it. But likely this will require attention. As for the AutoDrive controller, we would recommend very gentle curves to begin with.

When having a longer curved line (> 100 points), the Display must stream the coordinates to AgJunction:

- Up to 100 points can be given at a time
- It must overlap with at least 3 points from the previous set – thus including 3 points we already passed



6. Engage / Run Command & Tracking

If the AutoDrive system is active, the machine will only drive & steer (automatically) while the "RunCommand" is active. The RunCommand is allowed to become true if:

- GPS PPP correction is available & Machine allows AutoDrive (see DSSTAT message)
- System was activated before
- System streamed at least 100 points

So when switching on the contact or entering a field, it may take a while before the Run command can be given. Note at this point, the machine might be off the intended line. For a first try, it is recommended to put the machine right in front of the starting point. AgJunction can steer towards the line to some extend, but shouldn't start too far away.

Once the "RunCommand" turns on, the machine will drive forward (or in reverse if the next point says so) slowly (1 kph), as AgJunction cannot activate without driving first. After a short delay, the AutoSteer Engage request will be sent towards AgJunction. When the AutoSteer system is actually engaged, it will report back (see right message, "AutoSteer Engaged"):

ADJOB	**AutoDrive Job**							
(J1939) Message AutoDrive System to Display, containing job state								
PGN	65484	0xFFCC			**CAN BUS**	bus 2 on Main Display		
source	29 ?				**Priority**	6		
Transmit ra	Every 1s or when state changed				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1								
Byte2		Error Code (0 = ok)			RunCommand		SystemActive	
Byte3		Current waypoint index (i.e. #1403 out of #5000) - i.e. progress						
Byte4								
Byte5		Line total point count						
Byte6								
Byte7		Job ID (0..65530)						
Byte8								

DSSTAT	**DirectSteer Status**							
(J1939) Message from Display to AutoDrive System, containing current state								
PGN	65482	0xFFCA			**CAN BUS**	bus 2 on Main Display		
source	40				**Priority**	6		
Transmit ra	Every 200 msec				**Length**	8 bytes		
	Bit 8	Bit 7	Bit 6	Bit 5	Bit 4	Bit 3	Bit 2	Bit 1
Byte1	GPS PPP available		AutoSteer Engaged		Header Down		Current Direction	
Byte2	AutoSteer interrupt/reject reason						AutoDrive allowed	
Byte3		Perpendicular distance to line cm					1 cm per bit	
Byte4							offset -1000 cm	
Byte5		Overlap setting cm					offset -125 cm	
Byte6		Picker Reel width (shaft-to-shaft) cm					1 cm per bit	
Byte7		lhTipDistance cm					no offset	
Byte8					1 cm per bit, no offset			

This means that:

- The first meter(s), the machine has no waypoint yet, it will simply drive forward
    - This can be problematic if the machine was halted during a curve !
- The feedback “AutoSteer Engaged = TRUE” status might be a few seconds after the RunCommand was given

If the autosteer refuses to engage for whatever reason, the machine will halt and report an “AutoSteer interrupt/reject” code (which resembles a certain reason such as manually overrided, bad GPS signal, cannot acquire line, et cetera). If this happens, the system can try again by resetting the RunCommand, and otherwise an operator is needed.

During the ride, the AutoDrive system can also ask the machine to halt by turning off the RunCommand – for example if the machine needs to unload or receive a line update first. Resume by turning on the RunCommand again – but be aware that the same initial meters & delay are applied as written above; it may pass a few waypoints before it actually engaged & it might be slightly off the line by then!

The AutoDrive system can report the current waypoint index (“progress”) back with the ADJOB message. Also if the AutoDrive system stopped working for whatever reason, it can use this message to raise an error. Once again, the machine will halt if the errorCode field is not 0.